



AI Literacy

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Summary

In this session, students have learned about:

- What AI Literacy is
- What AI system is
- Real-world applications of AI
- Impacts of AI
- Ethical AI

AI Users in Indonesia

Katadata Insight Center shows that 64.7% of respondents in Indonesia have used AI technology, especially to search for information, shop online and edit media.

- 77.8% of them says they use AI every day.
- ChatGPT is used by approx 85% of AI users in Indonesia.
- Meta AI was reported to be used by 72% of respondents, and Google Gemini by 65%.

Source: <https://www.graphie.co.id/>

TEKNOLOGI

Adopsi AI Generatif Melesat, Literasi Digital Indonesia Masih Tertinggal

Skor Indeks Masyarakat Digital Indonesia untuk pilar literasi digital pada 2025 berada di kisaran 44-49 poin, yang berarti masih masuk dalam kategori sedang.

13 Feb 2026 19:03 WIB · Ekonomi & Bisnis



AI Literacy

AI literacy refers to a foundational conceptual understanding of AI. It focuses on knowledge, critical thinking, and ethical awareness rather than technical skill.

Individuals expected can explain:

- What AI systems are (e.g., machine learning, neural networks)
- Identify their real-world applications (like recommendation engines or facial recognition)
- Discuss societal implications such as bias, privacy risks, or economic disruption.
- Able to design, manage, engage, and create with AI
- Developing the ability to critically evaluate AI output

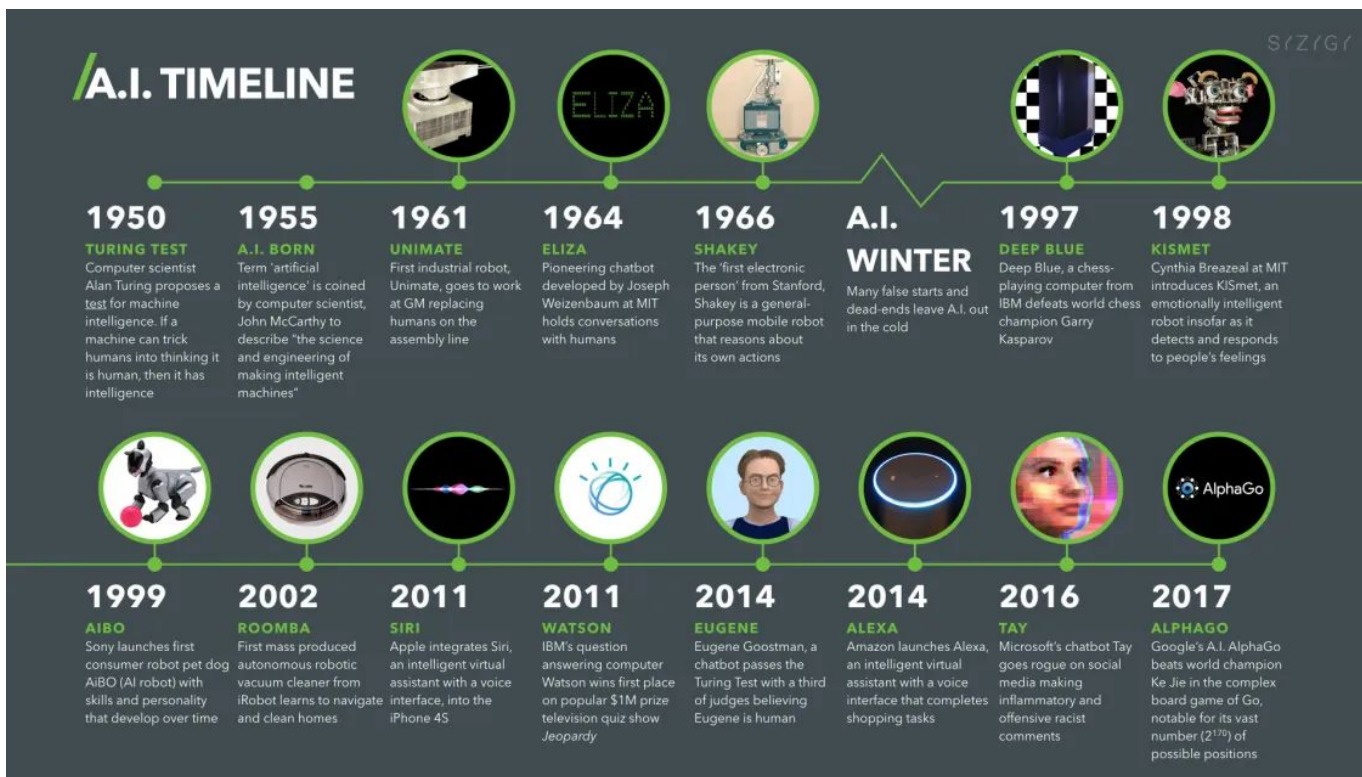


What is AI?

Raise your hand, if ...

- You have used AI before
- You are excited and interested by the potential of AI
- You are skeptical about the use of AI

A Brief History of AI



Understanding AI

A particular computer system or machine that has some of the qualities that the human brain has, such as the ability to interpret and produce language in a way that seems human, recognize or create images, solve problems, and learn from data supplied to it
(Cambridge Dictionary)

$$f(x) = y$$

AI is not human. AI systems use algorithms to combine step-by-step procedures with statistical inferences (e.g. weights and biases) to process data, detect patterns and generate outputs based on probabilities (Russell & Norvig, 2022).

1950's



Artificial intelligence (AI)

Human intelligence exhibited by machines

1980's



Machine learning

AI systems that learn from historical data

2010's



Deep learning

Machine learning models that mimic human brain function

2020's



Generative AI (Gen AI)

Deep learning models (foundation models) that create original content

Machine Learning

Involves creating models by training an algorithm to make predictions or decisions based on data.

Machine learning techniques or algorithms, including

- linear regression,
- logistic regression,
- decision trees,
- random forest,
- support vector machines (SVMs),
- k-nearest neighbor (KNN)
- clustering
- (Artificial) neural network

Deep Learning

Deep learning is a subset of machine learning that uses multilayered neural networks, called deep neural networks, that more closely simulate the complex decision-making power of the human brain.

These multiple layers enable unsupervised learning: they can automate the extraction of features from large, unlabeled and unstructured data sets, and make their own predictions about what the data represents.

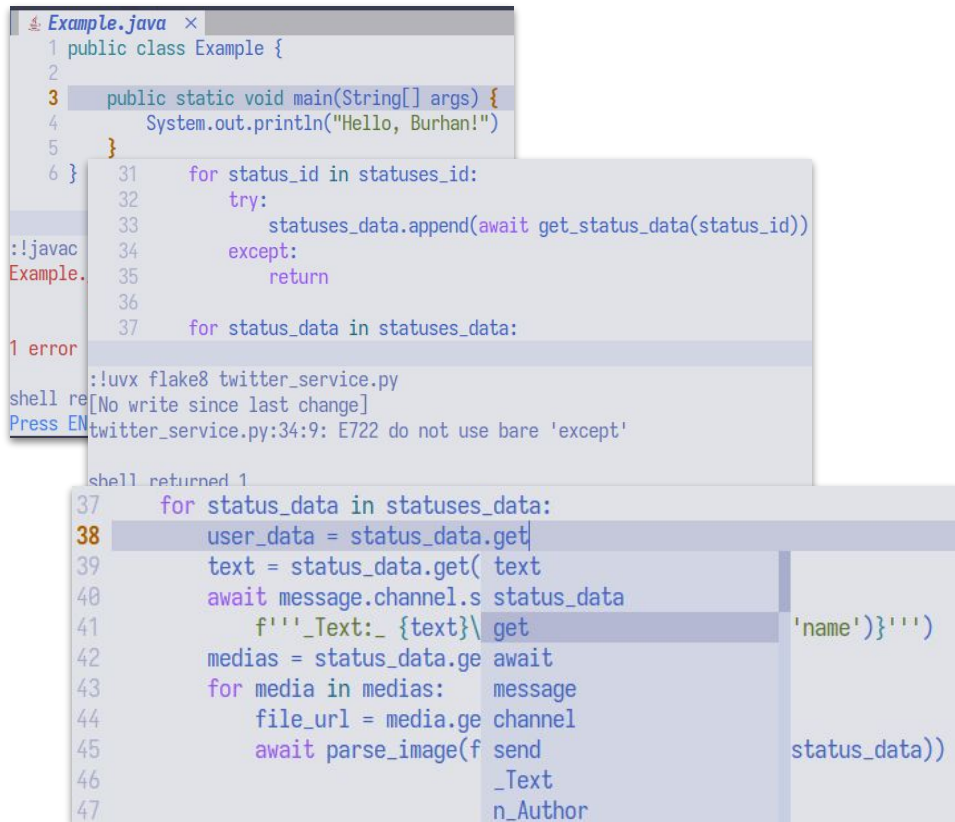
Generative AI

Generative AI, sometimes called "gen AI", refers to deep learning models that can create complex original content such as long-form text, high-quality images, realistic video or audio and more in response to a user's prompt or request.

AI Applications

There are many types of AI systems, which operate differently depending on their purpose, programming and training data (Burrell, 2016; Russell & Norvig, 2022). Depending on the type of tool, users may know they are interacting with AI; at other times, AI systems may operate in the background or as part of a tool and its influence is not evident (Bender et al., 2021).

AI in Programming (Before ChatGPT)



The screenshot shows a code editor with two files. The top file, 'Example.java', contains a Java class with a main method and a loop. The bottom file, 'twitter_service.py', contains Python code for processing Twitter data. A red error message is displayed at the bottom of the editor, indicating a syntax error in the Python code.

```
Example.java
1 public class Example {
2
3     public static void main(String[] args) {
4         System.out.println("Hello, Burhan!")
5     }
6 }

31 for status_id in statuses_id:
32     try:
33         statuses_data.append(await get_status_data(status_id))
34     except:
35         return
36
37 for status_data in statuses_data:
38     user_data = status_data.get('user')
39     text = status_data.get('text')
40     await message.channel.send(status_data['text'])
41     f'_{Text}_{text}_{user_data.get('name')}'
42     medias = status_data.get('media')
43     for media in medias:
44         file_url = media.get('url')
45         await parse_image(file_url, status_data)
46         _Text
47         n_Author
```

```
shell re[No write since last change]
Press EN
twitter_service.py:34:9: E722 do not use bare 'except'
```

```
shell returned 1
```

- “AI” has been commonly used in programming since 0 BC (Before ChatGPT)
- Examples:
 - Compiler (e.g., **javac**)
 - Static analysis (e.g., **flake8**, **ruff** in Python)
 - Code completion (e.g., IntelliSense, Ctrl-Space shortcut when typing code on IDE)
- The examples above use classical AI approach such as rule-based AI
 - Essentially, conditionals (if-else) and pattern matching
 - Rule-based AI still has its use even today in assisting programmer
- As mentioned in Brief History of AI, AI is not that “new” in our field!
 - It’s already been here since 1950
 - Big data-, statistic-based AI approach is the one currently popular right now

AI in Programming

```
8 def item_list(request):  
9     items = Item.objects.all()
```

← SORT ITEM LIST BY NAME

Start Over

views.py

Make the item list at `@file:views.py` to be sorted by its name in ascending order

6:00 PM

I'll verify the current queryset in `views.py`, then make the smallest ordering change and run a focused Django check.

Read `views.py`

The list is currently ordered by newest `created_at`, so the controlling change is the `item_list` queryset. I'm replacing that ordering with `name` ascending; Django's default ascending order is expressed without a leading minus.

The queryset now uses `order_by('name')`. I'll run Django's focused system check to confirm the edit leaves the project valid.

- Text-based generative AI is a next token “predictor”
 - Emphasise on **predictor** → based on probability
 - Similar to the autocompletion used by keyboard app in your phone
- More advanced AI models have been trained to simulate “thinking” and to be able to invoke “tool calls”
- It does help in assisting our work as ~~programmer~~ software engineer
 - Why the programmer word is struck?
- Many programming tasks are **repetitive** or following certain **code writing patterns**
 - Can be automated using AI, classic and/or generative
 - Look up about model-driven development in Software Engineering course as an example of classic AI approach in generating code
- Since the repetitive parts in programming can be “automated”, software development can focus more on the “engineering” aspect
 - E.g., requirements gathering, UI prototyping, system analysis & design
 - Even when some argue those engineering activities can be automated as well, it is still important to know the fundamentals to actually understand and verify whatever the AI has produced!

AI in Software Development

Key areas where AI is used in software development include:

- Code generation: Github Copilot, TabNine
- Automated Testing and Bug Detection: DeepCode, Snyk
- Project management
- Documentation
- Refactoring and optimization: [Refactoring.ai](https://refactoring.ai), JetBrains IntelliJ IDEA
- Security enhancement
- DevOps and CI/CD pipelines
- UX design
- Architecture design

Key to Consider

- Developing and maintaining AI relies on humans to design algorithms, collect, manage, evaluate, and label data
- AI is trained on vast datasets sourced from publicly available information
- AI systems can gather data during interactions with users
- AI systems are trained to identify patterns among data elements

Student Group Activity (4 students/group)

- Use your search engine of choice to find information and articles about AI
- Combine your search about AI with something that you are interested in
- Discuss with your group the information you have found and consider the following:
 - How is AI being presented?
 - How do you feel about what you have read? why?



Impact of AI

Woman Marries ChatGPT AI Character After He Proposed

AI marriage in Japan and the future symbol of love

BY METROPOLIS · JULY 3, 2026

Yurina Noguchi recently held a wedding ceremony with an AI partner she created using ChatGPT, after what she describes as a gradual emotional relationship that culminated in a proposal from him.

While legally binding i
forming long-term, er
hologram devices and

AI firm wins high court ruling after photo agency's copyright claim

Ruling in case brought by Getty Images against Stability AI is seen as a blow to copyright owners

Impact of AI in Software Engineering

The Register

SECURITY

Curl shuts bug bounty program to remove incentive for submitting AI slop

Maintainer hopes hackers send bug reports anyway, will keep shaming 'silly' ones

Simon Sharwood

10 

Wed 21 Jan 2026 // 05:29 UTC

[Curl shuts bug bounty program to stop AI slop](#)

Sep 9, 2025

AI-Generated Code: A Double-Edged Sword for Developers

By Natalie Tischler

If you think AI-generated code is saving time and boosting productivity, you're right. But here's the problem: it's also introducing security vulnerabilities at an alarming rate. [Our latest research reveals](#) that **45% of AI-generated code contains security flaws**, turning what should be a productivity breakthrough into a potential security nightmare.

AI coding tools like [GitHub Copilot](#), [Claude Code](#), and [ChatGPT](#) have revolutionized software development, enabling developers to generate functional applications with simple prompts. This shift toward "vibe coding" — trusting AI to handle implementation while focusing on ideas — has democratized programming and accelerated development cycles. However, this speed comes at a cost: security vulnerabilities that can compromise entire applications and organizations.

Ethical or Not Ethical? Activity

- Get into groups of 2-4 people.
- In this activity, you will read cards with different examples of a hypothetical usage for AI. You will sort those cards into buckets labelled "Ethical" or "Not Ethical."
- Discuss with your group if there is any circumstance wherein a usage could be altered in order to become ethical or not ethical.

Let's Discuss

- How did you determine if it was considered ethical or not?
- What were the common concerns of your group?
- Going forward, has this exercise changed how you will approach prompting with AI or consuming media?

Potential Risk of AI in Software Engineering

AI brings significant advantages to software development but it also presents potential risks that must be proactively managed.

- **Bias in AI models:** Regularly auditing AI outputs for fairness and integrating bias detection tools can also help ensure more equitable outcomes.
- **Overreliance on AI:** Developers can become overly dependent on AI tools for coding, debugging or testing, which might lead to a decline in their fundamental programming skills. To counter overreliance, developers should use AI as an assistive tool while also maintaining and honing their own technical expertise.
- **Security vulnerabilities:** AI-generated code can introduce security vulnerabilities if not properly vetted. human oversight should remain a critical component of code review.
- **Lack of transparency:** Many AI models, particularly in machine learning, operate in ways that are not entirely transparent to users. To improve transparency, developers should use more interpretable models whenever possible



Vibe Coding?

Tips LLM Prompting for Programming

- Remember that, AI is not human! They need context
- LLM prompts are not like search engine - a short prompt is not helpful.
- Be clear and fully describing the problem you need to solve. Describe the goal of the program. Give it specific information (eg: using library ...)
- You need to provide multiple prompts to get to the solution

Evaluation AI Output

Using AI responsibly requires more than knowing which tools to use. It requires the ability to **think critically** about AI-generated information, understand where it comes from, recognise its limitations and make informed decisions based on human judgement.

- **Authority:** Are there references to where the information came from? Are the citations real?
- **Bias:** What perspective is the source approaching the subject from? Does it acknowledge that perspective?
- **Credibility:** Are there any claims or information that have not been backed up with evidence?
- **Currency:** How up to date is the data your tool has been trained on? Is that information available?
- **Relevance:** Is the generated output relevant to your work? How might you adjust the prompt to get the information you want?

Reflect!

- Which AI model??
- What is the initial prompt and what is the model's response
- Were you able to integrate the model's code responses into your own code and get it to compile successfully?
- If you needed to fix bugs, what was your process? What was easy/hard about that process? Did you simply ask the model to fix its errors, or did you attempt to fix them in the code yourself?
- Please reflect on which of the prompts that you used generated the most useful responses, and reflect on if there was anything specific that you included that might have helped get good responses.

References

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3. <https://researchguides.library.vanderbilt.edu/c.php?g=1443398&p=10722153>
4. <https://web.stanford.edu/class/archive/cs/cs106b/cs106b.1262/assignments/7-huffman-ai-coding/>
5. https://researchguides.library.vanderbilt.edu/ld.php?content_id=81139914
6. <https://craft.stanford.edu/resources/> (More AI Literacy Activities)
7. Geeks for Geeks